① Encontre o raio de convergência

* Usar o teste da razão ou teste da integral. Com o resultado avaliar $|x|<1$, para determinar o espaço de convergência

1) $\sum_{n=0}^{\infty} (-1)^n \frac{x^n}{n+1}$ * Teste da razão: $\lim_{n\to\infty} \left| \frac{\frac{\cancel{x^n}.x}{n+2}}{\frac{\cancel{x^n}}{n+1}} \right| = \lim_{n\to\infty} \left| \frac{x.(n+1)}{n+2)} \right| = |x|.\lim_{n\to\infty} \left| \frac{n+1}{n+2} \right| = |x|$ * Raio: $|x|<1 \rightarrow -1<x<1$ raio $= 1$.

2) $\sum_{n=0}^{\infty} (4x)^n$ * Teste da razão: $\lim_{n\to\infty} \left| \frac{\cancel{(4x)^n}.(4x)}{\cancel{(4x)^n}} \right| = |4x|$ * Validar intervalo: $-1 < 4x < 1 \Rightarrow -\frac{1}{4} < x < \frac{1}{4}$ ∴ raio $= 1/4$

3) $\sum_{n=1}^{\infty} \frac{(2x)^n}{n^2}$ * Teste da razão: $\lim_{n\to\infty} \left| \frac{\frac{\cancel{(2x)^n}.(2x)}{(n+1)^2}}{\frac{\cancel{(2x)^n}}{n^2}} \right| = \lim_{n\to\infty} \left| \frac{2x.n^2}{(n+1)^2} \right| = |2x|.\lim_{n\to\infty} \left| \frac{n^2}{(n+1)^2} \right| = |2x|.\lim_{n\to\infty} \left| \cancel{\frac{n^2}{n^2+2n+1}} \xrightarrow{} 1 \right|$ ∴ raio $= 1/2$

5) $\sum_{n=0}^{\infty} \frac{(2x)^n}{n!}$ * Teste da razão: $\lim_{n\to\infty} \left| \frac{\frac{\cancel{(2x)^n}.(2x)}{(n+1).\cancel{n!}}}{\frac{\cancel{(2x)^n}}{\cancel{n!}}} \right| = \lim_{n\to\infty} \left| \frac{(2x)}{(n+1)} \right| = |2x|.\lim_{n\to\infty} \left| \frac{1}{n+1} \right| \rightarrow 0 = 0$ * Converge para todo Real: $R=\infty$

② Encontre o intervalo de convergência:

7) $\sum_{n=0}^{\infty} \left(\frac{x}{2}\right)^n$ * Teste da raiz: $\lim_{n\to\infty} \sqrt[n]{\left(\frac{x}{2}\right)^n} = \frac{|x|}{2} \Rightarrow -2<x<2$

Se $x=2 \Rightarrow \sum_{n=1}^{\infty} \left(\frac{2}{2}\right)^n = \sum_{n=1}^{\infty} (1)^n$ Diverge.

Se $x=-2 \Rightarrow \sum_{n=1}^{\infty} \left(\frac{-2}{2}\right)^n = \sum_{n=1}^{\infty} (-1)^n$ Diverge.

∴ Intervalo é $(-2,2)$

9) $\sum_{n=1}^{\infty} \frac{(-1)^n x^n}{n}$ * Teste $\lim_{n\to\infty} \frac{\frac{\cancel{(-1)^n}.(-1).\cancel{x^n}.x}{n}}{\frac{\cancel{(-1)^n}.\cancel{x^n}}{n}} = \lim_{n\to\infty} |(-1).x| = |x| \rightarrow -1<x<1$

Se $x=-1 \Rightarrow \sum_{n=1}^{\infty} \frac{(-1)^n.(-1)^n}{n} = \sum_{n=1}^{\infty} \frac{(-1)^{2n}}{n} = 1+\frac{1}{4}+\frac{1}{6}\dots$ Diverge

Se $x=1 \Rightarrow \sum_{n=1}^{\infty} \frac{(-1)^n.1^n}{n} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ $\lim_{n\to\infty} = 0$ $\frac{1}{n+1} < \frac{1}{n}$ converge

∴ Intervalo: $(-1,1]$

11) $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ * $\lim_{n\to\infty} \left| \frac{\frac{\cancel{x^n}.x}{(n+1).\cancel{n!}}}{\frac{\cancel{x^n}}{\cancel{n!}}} \right| = \lim_{n\to\infty} \frac{|x|}{(n+1)} = |x|.0 = 0$ Raio infinito de convergência ∴ $(-\infty,\infty)$

13) $\sum_{n=0}^{\infty} (2n)! \left(\frac{x}{2}\right)^n$ * $\lim_{n\to\infty} \left| \frac{(2.(n+1))!\,\cancel{\left(\frac{x}{2}\right)^n}.\left(\frac{x}{2}\right)}{(2n)!\,\cancel{\left(\frac{x}{2}\right)^n}} \right| = \frac{|x|}{2}.\lim_{n\to\infty} \frac{(2.(n+1))(2n)!}{(2n)!} = \infty$ ∴ Só converge no centro: $[0]$

15) $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}.x^n}{4^n}$ * $\lim_{n\to\infty} \left| \frac{\frac{\cancel{(-1)^n}.(-1)^2.\cancel{x^n}.x}{\cancel{4^n}.4}}{\frac{\cancel{(-1)^n}.(-1).\cancel{x^n}}{\cancel{4^n}}} \right| = \lim_{n\to\infty} x\frac{1}{4}. x\frac{1}{4} \rightarrow -4<x<4$

Se $x=-4$ $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}.(-4)^n}{4^n} = \sum_{n=1}^{\infty} \frac{(-1)^n.(-1).(-1)^n.4^n}{4^n} = \sum_{n=1}^{\infty} (-1)^{2n+1}$ Diverge

Se $x=4$ $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}.4^n}{4^n} = \sum_{n=1}^{\infty} (-1)^{n+1}$ Diverge.

∴ Intervalo: $(-4,4)$

17) $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}.(x-5)^n}{n.5^n}$ * $\lim_{n\to\infty} \left| \frac{\frac{\cancel{(-1)^n}.(-1)^2.\cancel{(x-5)^n}.(x-5)}{(n+1).\cancel{5^n}.5}}{\frac{\cancel{(-1)^n}.(-1).\cancel{(x-5)^n}}{n.\cancel{5^n}}} \right| = \lim_{n\to\infty} \left| \frac{(x-5).n}{-5.(n+1)} \right| = \frac{1}{5}\lim_{n\to\infty} \frac{(x-5).\cancel{n}}{\cancel{n}.(1+\frac{1}{n})} = \frac{(x-5)}{5}.\lim_{n\to\infty} 1 \rightarrow -1 < \frac{(x-5)}{5} < 1 \rightarrow -5 < x-5 < 5 \rightarrow 0<x<10$

Se $x=0$ $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}.(-5)^n}{n.5^n} = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}.(-1)^n.\cancel{(5)^n}}{n.\cancel{5^n}} = \sum_{n=1}^{\infty} \frac{(-1)^{2n+1}}{n} = \sum_{n=1}^{\infty} \frac{-1}{n}$ Diverge

Se $x=10$ $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}.\cancel{5^n}}{n.\cancel{5^n}} = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ $\lim = 0$ $a_{n+1} \leq a_n$ CONVERGE ∴ $(0,10]$

19) $\sum_{n=0}^{\infty} \frac{(-1)^{n+1}.(x-1)^{n+1}}{n+1}$ * $\lim_{n\to\infty} \left| \frac{\frac{\cancel{(-1)^n}.(-1)^2.\cancel{(x-1)^n}.(x-1)^2}{n+2}}{\frac{\cancel{(-1)^n}.(-1).\cancel{(x-1)^n}.\cancel{(x-1)}}{n+1}} \right| = \lim_{n\to\infty} \left| \frac{(x-1)\ (n+1)}{(n+2)} \right| = |(x-1)|.1 \Rightarrow \boxed{0<x<2}$

Se $x=0$ $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}.(-1)^{n+1}}{n+1} = \sum_{n=1}^{\infty} \frac{(-1)^{2n+2}}{n+1} = \sum_{n=1}^{\infty} \frac{1}{n+1}$ Diverge

Se $x=2$ $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}.\cancel{(1)^{n+1}}\rightarrow 1}{n+1} = \sum_{n=1} \frac{(-1)^{n+1}}{n+1}$ $\lim = 0$ $\boxed{a_{n+1} \leq a_n}$ CONVERGE

∴ $(0,2]$

21) $\sum_{n=1}^{\infty} \left(\frac{x-c}{c}\right)^{n-1} \rightarrow$ Geométrica $r = \frac{x-c}{c}$, converge se $r<1 \Rightarrow -1 < \frac{x-c}{c} < 1 \Rightarrow \boxed{0<x<2c}$

Se $x=0 \Rightarrow \sum_{n=1}^{\infty} (-1)^{n-1}$ Diverge

Se $x=2c \Rightarrow \sum_{n=1}^{\infty} (1)^{n-1}$ Diverge

∴ $(0,2c)$

23) $\sum_{n=1}^{\infty} \frac{n}{n+1}.(-2x)^{n-1}$ * $\lim_{n\to\infty} \left| \frac{\frac{(n+1).\cancel{(-2x)^n}}{n+2}}{\frac{n.\cancel{(-2x)^n}}{(n+1).(-2x)}} \right| = \lim_{n\to\infty} \left| \frac{(n+1).(n+1).(-2x)}{n+2)} \right| = \lim_{n\to\infty} \left| (-2x).\frac{n^2+2n+1}{n+2} \right| = 2x$ $\boxed{-\frac{1}{2}<x<\frac{1}{2}}$

Se $x=\frac{1}{2}$

Se $x=-\frac{1}{2}$

ambos divergem $(-\frac{1}{2}, \frac{1}{2})$

25) $\sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$ * $\lim_{n\to\infty} \left| \frac{\frac{x^{2n+3}}{(2n+3)!}}{\frac{x^{2n+1}}{(2n+1)!}} \right| = \lim_{n\to\infty} \left| \frac{\cancel{x^{2n}}.x^3.(2n+1)!}{\cancel{x^{2n}}.x.(2n+3)!} \right| = \lim_{n\to\infty} \left| x^2.\frac{\cancel{(2n+1)!}}{(2n+3).(2n+2).\cancel{(2n+1)!}} \right| = \infty$ ∴ Raio infinito

Forma de uma série de potência: $\sum_{n=0}^{\infty} a.r^n = \frac{a}{1-r}$, $|r|<1$

③ 1) $f(x) = \frac{1}{2-x}$, $c=0$ $\Rightarrow f(x) = \frac{1/2}{1-\frac{x}{2}}$ $\Rightarrow \sum_{n=0}^{\infty} \frac{1}{2}\cdot\left(\frac{x}{2}\right)^n$ e converge em $\boxed{-2<x<2} = (-2,2)$

3) $f(x) = \frac{1}{2-x}$, $c=5 \Rightarrow f(x) = \frac{1}{2-x} = \frac{1}{2-(x-5+5)} = \frac{1}{-3-(x-5)} = \frac{-1/3}{1-\left[-1.\frac{x-5}{3}\right]} \Rightarrow f(x) = \sum_{n=0}^{\infty} -\frac{1}{3}\cdot\left(-1.\frac{x-5}{3}\right)^n = \sum_{n=0}^{\infty} -\frac{1}{3}\cdot(-1)^n\cdot\left(\frac{x-5}{3}\right)^n$ e converge em $\boxed{2<x<8} = (2,8)$

5) $f(x) = \frac{3}{2x-1} = \frac{3}{-1+2x} = \frac{-3}{1-2x} \Rightarrow f(x) = \sum_{n=0}^{\infty} -3.(2x)^n$ e converge $\boxed{-\frac{1}{2}<x<\frac{1}{2}} = \left(-\frac{1}{2},\frac{1}{2}\right)$

7) $f(x) = \frac{1}{2x-5} = \frac{1}{-5+2x} = \frac{-1/5}{1-\frac{2}{5}x} \Rightarrow f(x) = \sum_{n=0}^{\infty} -\frac{1}{5}\left(\frac{2}{5}x\right)^n$ converge $\boxed{\frac{-5}{2}<x<\frac{5}{2}} = \left(-\frac{5}{2},\frac{5}{2}\right)$ * Centrado em 0.

$c=-3$

$f(x) = \frac{1}{2x-5} = \frac{1}{-5+2(x+3-3)} = \frac{1}{-11+2.(x+3)} = \frac{-1/11}{1-\frac{2}{11}(x+3)} \Rightarrow f(x) = \sum_{n=0}^{\infty} \frac{-1}{11}\cdot\left(-\frac{2}{11}.(x+3)\right)^n$ converge $\boxed{\frac{-17}{2}<x<\frac{5}{2}} = \left(-\frac{17}{2},\frac{5}{2}\right)$

9) $f(x) = \frac{3}{x+2} = \frac{3}{2+x} = \frac{3/2}{1-\left(-\frac{x}{2}\right)} \Rightarrow f(x) = \sum_{n=0}^{\infty} \frac{3}{2}\cdot\left(-\frac{x}{2}\right)^n$ converge em $\boxed{-2<x<2} = (-2,2)$

11) $f(x) = \frac{3x}{x^2+x-2} \Rightarrow f(x) = \frac{1}{(x-1)} + \frac{2}{(x+2)} = \frac{-1}{1-x} + \frac{1}{1-\left(-\frac{x}{2}\right)} = \sum_{n=0}^{\infty}(-1).(x)^n + \sum_{n=0}^{\infty}(-1)^n\left(\frac{x}{2}\right)^n$

I II

Testando intervalos:

I $\sum_{n=0}^{\infty}(-1).(x)^n$ Diverge II $\sum_{n=0}^{\infty}(-1)^n\cdot\left(\frac{x}{2}\right)^n$

$\boxed{-1<x<1}$ $\boxed{-2<x<2}$

* Frações Parciais: $x^2+x-2 = (x+2).(x-1)$

$\frac{3x}{(x+2).(x-1)} = \frac{A}{(x+2)} + \frac{B}{(x-1)} = \frac{A.(x-1)+B.(x+2)}{(x+2).(x-1)} \Rightarrow A.(x-1)+B(x+2) = 3x$

Se $x=1$: $B+2B=3 \Rightarrow B=1$

Se $x=-2$: $-3A=-6 \Rightarrow A=2$

$\therefore f(x) = \frac{1}{(x-1)} + \frac{2}{(x+2)}$ * Agora vai abrir em duas séries.

$\therefore$ O intervalo de convergência que são compatível é o intervalo $(-1,1)$.

13) $f(x) = \frac{2}{1-x^2} = \sum_{n=0}^{\infty} 2.x^{2n} \Rightarrow$ Intervalo: $|x^2|<1 \Rightarrow |x|<1 \Rightarrow \boxed{-1<x<1} = (-1,1)$

④ $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!}.(x-c)^n$

1) $f(x) = e^{2x}$, $c=0$

$f(x) = e^{2x} \Rightarrow f(0) = 1$

$f'(x) = 2e^{2x} \Rightarrow f'(0) = 2$

$f''(x) = 4e^{2x} \Rightarrow f''(0) = 4$

$f'''(x) = 8e^{2x} \Rightarrow f'''(0) = 8$

$\vdots$

$P(x) = \frac{1}{1!}.(x)^n + \frac{2.(x)^n}{2!} + \frac{4.(x)^n}{3!} + \frac{8.(x)^n}{4!}$

$\boxed{P(x) = \sum_{n=0}^{\infty} \frac{2^n}{n!}.(x)^n}$

3) $f(x) = \cos(x)$ $c = \frac{\pi}{4}$

$f\left(\frac{\pi}{4}\right) = \cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$

$f'\left(\frac{\pi}{4}\right) = -\operatorname{sen}\left(\frac{\pi}{4}\right) = -\frac{\sqrt{2}}{2}$

$f''\left(\frac{\pi}{4}\right) = -\cos\left(\frac{\pi}{4}\right) = -\frac{\sqrt{2}}{2}$

$f'''\left(\frac{\pi}{4}\right) = \operatorname{sen}\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$

$f^{IV}\left(\frac{\pi}{4}\right)$

$P(x) = \sum_{n=0}^{\infty} \frac{\frac{\sqrt{2}}{2}}{0!}.\left(x-\frac{\pi}{4}\right)^0 - \frac{\frac{\sqrt{2}}{2}}{1!}.\left(x-\frac{\pi}{4}\right)^1 - \frac{\frac{\sqrt{2}}{2}}{2!}.\left(x-\frac{\pi}{4}\right)^2 + \frac{\frac{\sqrt{2}}{2}}{3!}.\left(x-\frac{\pi}{4}\right)^3 + \ldots$

$\boxed{P(x) = \sum_{n=0}^{\infty} \frac{\sqrt{2}}{2}\cdot\left(\frac{(-1)^{n(n+1)/2}}{n!}\right)\cdot\left(x-\frac{\pi}{4}\right)^n}$

5) $f(x) = \ln(x)$, $c=1$

$f(1) = \ln 1 \Rightarrow f(1) = 0$

$f'(x) = \frac{1}{x} \Rightarrow f'(1) = 1$

$f''(x) = -\frac{1}{x^2} \Rightarrow f''(1) = -1$

$f'''(x) = \frac{2}{x^3} \Rightarrow f'''(1) = 2$

$f^{IV}(x) = \frac{-6}{x^4} \Rightarrow f^{IV}(1) = -6$

$P(x) = \frac{0}{0!}.(x-1)^n + \frac{1}{1!}.(x-1)^n + \frac{1}{2!}.(x-1)^n - \frac{2}{3!}.(x-1)^n - \frac{6}{4!}.(x-1)^n \ldots$

$P(x) = \sum_{n=0}^{\infty}$ * Como chegar nessa série?

7) $f(x) = \operatorname{sen}(2x)$, $c=0$

$f(x) = \operatorname{sen}(2x) \Rightarrow f(0) = 0$

$f'(x) = 2.\cos(2x) \Rightarrow f'(0) = 2$

$f''(x) = -4.\operatorname{sen}(2x) \Rightarrow f''(0) = 0$

$f'''(x) = -8.\cos(2x) \Rightarrow f'''(0) = -8$

$f^{IV}(x) = 16.\operatorname{sen}(2x) \Rightarrow f^{IV}(0) = 0$

$f^{V}(x) = 32.\cos(2x) \Rightarrow f^{V}(0) = 32$

* Só tem termo nos ímpares e sinal varia entre eles.

$P(x) = \frac{0}{0!}.(x)^n + \frac{2}{1!}.(x)^n + \frac{0}{2!}.(x)^n - \frac{8.(x)^n}{3!} + \frac{0}{4!}.(x)^n$

$\boxed{P(x) = \sum_{n=0}^{\infty} \frac{(-1)^n.(2x)^{2n+1}}{(2n+1)!}}$

$\boxed{P(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!}.(x-c)^n}$

7) $f(x) = \operatorname{sen}(2x)$

$P(x) = \frac{2}{2!}.(x) - \frac{8.(x)^3}{3!} + \frac{32.x^5}{5!}$

$P(x) = \frac{2^1}{1!}x - \frac{2^3.x^3}{3!} + \frac{2^5.x^5}{5!}$

$\boxed{P(x) = \sum_{n=0}^{\infty} (-1)^n \frac{(2.x)^{(2n+1)}}{(2n+1)!}}$

$n=0$ $f(0) = \operatorname{sen}(0) \Rightarrow f(0) = 0$

$n=1$ $f'(0) = 2.\cos(0) \Rightarrow f'(0) = 2$

$n=2$ $f''(0) = -4.\operatorname{sen}(0) \Rightarrow f''(0) = 0$

$n=3$ $f'''(0) = -8.\cos(0) \Rightarrow f'''(0) = -8$

$n=4$ $f^{IV}(0) = 16.\operatorname{sen}(0) \Rightarrow f^{IV}(0) = 0$

$n=5$ $f^{V}(0) = 32.\cos(0) \Rightarrow f^{V}(0) = 32$

$\vdots$

Série de Mallaurin: $\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} \cdot (x)^n$

5 11) $f(x) = e^{x^2/2}$, $c=0$

$u = x^2/2 \Rightarrow f(u) = e^u$

$n=0$ $f(u) = e^u \Rightarrow f(0) = 1$

$n=1$ $f'(u) = e^u \Rightarrow f(0) = 1$

$\vdots$

$\frac{1}{0!} \cdot u^0 + \frac{1}{1!} \cdot u^1 + \frac{1}{2!} \cdot u^2 \ldots$

$\rightarrow \frac{1}{0!} \cdot (x^2/2)^0 + \frac{1}{1!} \cdot (x^2/2)^1 + \frac{1}{2!} \cdot (x^2/2)^2 \ldots$

$1 + \frac{x^2/2}{1!} + \frac{x^4/2^2}{2!} + \ldots = \sum_{n=0}^{\infty} \frac{x^{2n}}{n! \cdot 2^n}$

$\therefore f(x) = e^{x^2/2} = \sum_{n=0}^{\infty} \frac{x^{2n}}{n! \cdot 2^n}$ //

6 a) $\int_0^x (e^{-t^2}-1)\,dt = \int_0^x e^{-t^2}dt - 1$

ou seja: $\int_0^x (e^{-t^2}-1)\,dt = \int_0^x \left(\left[\sum_{n=0}^{\infty} \frac{(-t^2)^n}{n!}\right] - 1\,dt\right) = \int_0^x \left[\sum_{n=0}^{\infty} \frac{(-t^2)^n}{n!}\right]dt - \int_0^x 1\,dt = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \cdot \left[\int_0^x (t)^{2n}dt\right] - [t]_0^x = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \cdot \left[\left[\frac{t^{2n+1}}{2n+1}\right]_0^x\right] - x =$

$= \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \cdot \left[\left[\frac{t^{2n+1}}{2n+1}\right]_0^x\right] - x = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \cdot \left[\frac{x^{2n+1}}{2n+1}\right] - x$

* $u = -t^2$

$e^{-t^2} = e^{u}$

$\boxed{\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n}$

* Expansão para $f(u) = e^{u}$

$n=0 \quad f(0) = e^0 \Rightarrow f(0) = 1$

$n=1 \quad f'(0) = e^0 \Rightarrow f'(0) = 1$

$\vdots$

$P(u) = \frac{f(0)}{0!} \cdot (u)^0 + \frac{f'(0) \cdot (u)^1}{1!} + \frac{f''(0) \cdot u^2}{2!} \ldots$

$\sum_{n=0}^{\infty} \frac{(u)^n}{n!} = \boxed{\sum_{n=0}^{\infty} \frac{(-t^2)^n}{n!}}$

* Jogada algébrica para remover aquele $-x$ da função, o 1º termo da soma é x:

$P(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \cdot \left[\frac{x^{2n+1}}{2n+1}\right] - x = \sum_{n=1}^{\infty} \frac{(-1)^n}{n!} \cdot \left[\frac{x^{2n+1}}{2n+1}\right] + x - x$

$\hookrightarrow \frac{(-1)^0}{0!} \cdot \left[\frac{x^{2 \cdot 0+1}}{2 \cdot 0+1}\right] = x$

$\boxed{\therefore P(x) = \sum_{n=1}^{\infty} \frac{(-1)^n \cdot x^{2n+1}}{n!\,(2n+1)}}$

* Será que é o mesmo caso?

c) $\int_0^x \frac{\operatorname{sen} t}{t}\,dt$

* Expansão Maclaurin para $\operatorname{sen}(t)$

$f(t) = \operatorname{sen}(t) \rightarrow f(0) = \operatorname{sen}(0) = 0$

$f'(t) = \cos(t) \rightarrow f'(0) = \cos(0) = 1$

$f''(t) = -\operatorname{sen} \rightarrow f''(0) = -\operatorname{sen}(0) = 0$

$f'''(t) = -\cos(t) \rightarrow f'''(0) = -\cos 0 = -1$

$\operatorname{sen}(t) = \sum \frac{f^{(n)}(0)}{n!} \cdot t^n$

$\operatorname{sen}(t) = \frac{0 \cdot t^0}{0!} + \frac{1 \cdot t^1}{1!} + \frac{0 \cdot t^2}{2!} + \frac{(-1) \cdot t^3}{3!} + \ldots = \frac{t^1}{1!} - \frac{t^3}{3!} + \frac{t^5}{5!} - \frac{t^7}{7!} + \ldots$

$\therefore \operatorname{sen}(x) = \sum_{n=0}^{\infty} \frac{(-1)^n \cdot t^{(2n+1)}}{(2n+1)!}$

$\Rightarrow \int_0^x \sum_{n=0}^{\infty} \frac{(-1)^n\, t^{(2n+1)}}{(2n+1)!} \cdot \frac{1}{t}\,dt = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \cdot \int_0^x t^{2n}dt = \sum_{n=0}^{\infty} \frac{(-1)^n \cdot (x)^{2n+1}}{(2n+1) \cdot (2n+1)!}$

$\therefore P(x) = \sum_{n=0}^{\infty} \frac{(-1)^n \cdot (x)^{2n+1}}{(2n+1) \cdot (2n+1)!} = \int_0^x \frac{\operatorname{sen} t}{t}\,dt$

e) $\int_0^x \frac{\ln(t+1)}{t}\,dt$

* Expansão de $\ln(t+1)$

$\boxed{\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} \cdot x^n}$

$n=0 \Rightarrow f(t) = \ln(t+1) \Rightarrow f(0) = 0$

$n=1 \Rightarrow f'(t) = \frac{1}{(t+1)} \Rightarrow f'(0) = 1$

$n=2 \Rightarrow f''(t) = -(t+1)^{-2} \Rightarrow f''(0) = -1$

$n=3 \Rightarrow f'''(t) = 2 \cdot (t+1)^{-3} \Rightarrow f'''(0) = 2$

$n=4 \Rightarrow f^{IV}(t) = -6 \cdot (t+1)^{-4} \Rightarrow f^{IV}(0) = -6$

$n=5 \Rightarrow f^{V}(t) = 24 \cdot (t+1)^{-5} \Rightarrow f^{V}(0) = 24$

$\frac{f^0(0)}{0!} \cdot x^0 + \frac{f^1(0)}{1!} \cdot x + \frac{f^2(0)}{2!} \cdot x^2 + \frac{f^3(0)}{3!} \cdot x^3 + \frac{f^4(0)}{4!} \cdot x^4 + \frac{f^5(0)}{5!} \cdot x^5$

$\frac{0}{0!} \cdot x^0 + \frac{1}{1!} \cdot x - \frac{1}{2!} \cdot x^2 + \frac{2}{3!} \cdot x^3 - \frac{6}{4!} \cdot x^4$

$\boxed{x = t}$

$\underset{n=0}{0} + \underset{n=1}{x^1} - \underset{n=2}{\frac{1}{2}x^2} + \frac{2}{3!} \cdot x^3 - \frac{6}{4!} \cdot x^4 + \ldots = \underset{n=1}{x} - \underset{n=2}{\frac{x^2}{2}} + \underset{n=3}{\frac{x^3}{3}} - \underset{n=4}{\frac{x^4}{4}} + \ldots \Rightarrow \sum_{n=1}^{\infty} \frac{(-1)^{(n-1)} \cdot x^n}{n}$

$\therefore \ln(t+1) = \sum_{n=1}^{\infty} \frac{(-1)^{(n-1)} \cdot t^n}{n}$

$\Rightarrow \int_0^x \frac{\ln(t+1)}{t}\,dt = \int_0^x \sum_{n=1}^{\infty} \frac{(-1)^{(n-1)} \cdot t^n}{n} \cdot \frac{1}{t}\,dt = \sum_{n=0}^{\infty} \frac{(-1)^{(n-1)}}{n} \cdot \int_0^x t^{(n-1)}dt = \sum_{n=0}^{\infty} \frac{(-1)^{(n-1)}}{n} \cdot \frac{t^n}{n} = \sum_{n=0}^{\infty} \frac{(-1)^{(n-1)} \cdot t^n}{n^2}$ //